VELLORE INSTITUTE OF TECHNOLOGY, VELLORE

B.Tech Computer Science and Engineering | First Year

Embedded Systems – Project Review

Smart AI-Based Voice Assistant for Elderly Assistance

An ESP32-S3 Based Smart Assistance System — Emergency SOS, Medicine Reminders and Wellness Check-ins

Mapped to the Embedded System Design Process

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Overview

Presentation Outline

Part A · Project Context

01 Problem statement & proposed solution

02 Why this is an embedded system

03 Elements of an embedded system in our device

04 The design process roadmap

05 Literature survey & gap addressed

Part B · Design Process Applied

Stage 1 Requirements — functional & non-functional

Stage 2 Specification — precise system behaviour

Stage 3 Architecture — hardware & software

Stage 4 Components — modules, tools & the 3 functions

Stage 5 System integration, testing & next steps

Motivation

Problem Statement

Independent elderly living becomes challenging when help, memory support, and caregiver communication are not immediately available.

1

Emergency Situations

Elderly users need a simple way to request urgent help without navigating a phone or app.

2

Medication Schedules

Missed or forgotten doses are common without timely, easy-to-understand reminders.

3

Caregiver Communication

Caregivers need remote notification of an SOS event, even with no cellular module fitted.

4

Emotional Wellbeing

Routine wellness check-ins are rarely built into conventional smart-assistant products.

Proposed Solution

An **ESP32-S3 based voice assistant** that combines conversational AI with three elderly-focused functions — **Emergency SOS notification**, **Medicine Reminders**, and **State-of-Mind wellness check-ins** — while traditional assistants mainly answer questions or perform general tasks without elderly-specific safety workflows.

Theory → Project

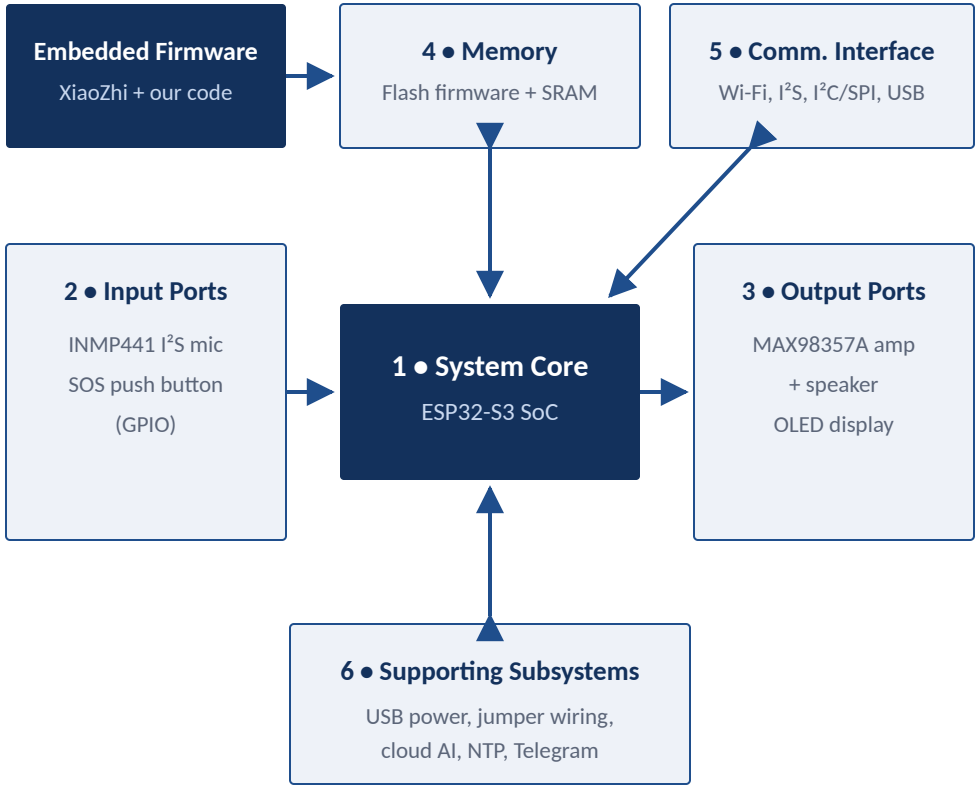
Why Our Device Is an Embedded System

An embedded system is an electronic / electro-mechanical system designed to perform a **specific function**, built from a combination of **hardware and firmware**. Our assistant fits this definition — it is not a general-purpose computer.

Criterion	General Purpose Computing System	Our Elderly Assistant (Embedded System)
Hardware & OS	Generic hardware + GPOS (Windows, Linux)	Special-purpose ESP32-S3 board running ESP-IDF firmware (FreeRTOS-based)
Applications	Alterable by the user	Fixed set: SOS, reminders, wellness check-ins, AI chat — firmware is pre-programmed
Selection factor	"Faster is better"	Application-specific: low cost, always-on, simple to use
Power	Not tailored for low power	Runs continuously from a USB supply; low-power SoC
Response time	Not time-critical	Time-critical: SOS alert and reminders must happen on time
Determinism	Need not be deterministic	Predictable SOS path: button → confirm → alert, every time

Theory → Project

Elements of an Embedded System in Our Device



Category of System Core

From the core categories (microprocessor, microcontroller, DSP, ASIC, PLD/FPGA, COTS), the ESP32-S3 is a **32-bit microcontroller / SoC**: CPU, RAM, flash interface, timers, GPIO, I²S, I²C and Wi-Fi radio on one chip — no external support chips needed for independent working.

Commercial Off-the-Shelf (COTS)

Like the TCP/IP plug-in module example, we use ready-made modules (INMP441, MAX98357A, OLED, dev board) and services (Telegram Bot API, NTP, XiaoZhi) — cheap, available, and they cut development time.

Microcontroller vs. microprocessor: a microprocessor would need external RAM, ROM, timers and I/O; choosing a microcontroller keeps the device small, low-cost and self-contained.

Methodology

The Embedded System Design Process — Applied

We follow the top-down design flow; estimates flow back (bottom-up refinement) whenever an assumption changes during implementation.

STAGE 1	STAGE 2	STAGE 3	STAGE 4	STAGE 5
Requirements	**Specification**	**Architecture**	**Components**	**Integration**
Informal description of what the user wants	Precise contract of system behaviour	Major hardware & software blocks	Modules and code designed & built	Assemble & test as a whole
Our project	**Our project**	**Our project**	**Our project**	**Our project**
Help in emergencies, remember medicines, check on wellbeing, notify caregiver; low cost & easy to use.	Inputs, outputs, data, user-interface behaviour and background actions for each of the 3 functions.	ESP32-S3 hardware block diagram + software flowchart and firmware task structure.	INMP441, MAX98357A, OLED, SOS button; XiaoZhi firmware, Telegram, NTP modules.	SOS path first, then reminders, then wellness — each tested end-to-end.

→ Top-down design

← Bottom-up refinement

Stage 1 · Requirements

Requirements Form

Captured using the standard requirements form (as in the GPS moving-map case study).

Field	Smart Elderly Voice Assistant
Name	Smart AI-Based Voice Assistant for Elderly Assistance
Purpose	A voice-first home device that lets an elderly user call for help, reminds them of medicines, and checks on their wellbeing — with caregiver notification over Wi-Fi.
Inputs	Voice via INMP441 I^2S microphone (continuous / on-demand); SOS push button (occasional, GPIO); network time from NTP (periodic)
Outputs	Speech & alert tones via MAX98357A + speaker; status text on OLED; Telegram SOS message to caregiver
Functions	Emergency SOS (button or voice keyword, with confirmation) • scheduled medicine reminders • periodic “State of Mind” check-ins • general AI conversation
Performance	SOS message sent within a few seconds of confirmation; reminders fire at the scheduled minute; natural, responsive voice replies
Manufacturing cost	Low-cost — built only from development-board-class COTS modules
Power	Always-on, powered from 5 V USB supply
Size / weight	Compact tabletop unit that fits beside a bed or chair

Stage 1 · Requirements

Functional & Non-Functional Requirements

Functional — what it must do

- Provide natural voice-based interaction
- Detect emergency requests through voice keywords
- Provide a physical SOS button for emergencies
- Send SOS notifications to a caregiver via Telegram
- Provide scheduled medicine reminders
- Perform periodic wellness / “State of Mind” check-ins
- Provide audio and visual feedback to the user

Non-functional — constraints on how well

Performance
Soft deadlines: SOS delivery in seconds, reminders on the scheduled minute (NTP-accurate clock).

Cost
Low manufacturing cost; low NRE by reusing open-source XiaoZhi firmware.

Size & Weight
Small tabletop device; no phone or app needed by the user.

Power
Mains-USB powered for always-on listening; low-power SoC.

Reliability
Confirmation step to avoid false alerts; hardware button works even if speech fails.

Usability
One button, simple spoken prompts, large OLED status for elderly users.

Stage 2 · Specification

System Specification

The specification is the precise contract between user needs and the architects — it says **what** the system does. Following the GPS example, it covers:

1

Data received

Digital audio samples from the I²S mic; button state on GPIO; current date/time from NTP server; AI replies from the XiaoZhi cloud service.

2

Stored data

Medicine name & reminder times stored in firmware configuration; Wi-Fi credentials; caregiver Telegram chat ID and bot token.

3

User interface

Speech in / speech out, one SOS button, OLED shows status (listening, SOS sent, medicine name). Every SOS needs spoken or button confirmation.

4

Operations on request

Answer general questions, raise SOS, give device info, respond to wellness answers (positive / neutral / concerning).

5

Background actions

Maintain Wi-Fi; periodic NTP re-sync; compare clock with next reminder; trigger scheduled check-ins.

SOS message (exact)

"SOS ALERT: Emergency assistance requested by the user. Please check on them immediately."

Stage 2 · Specification

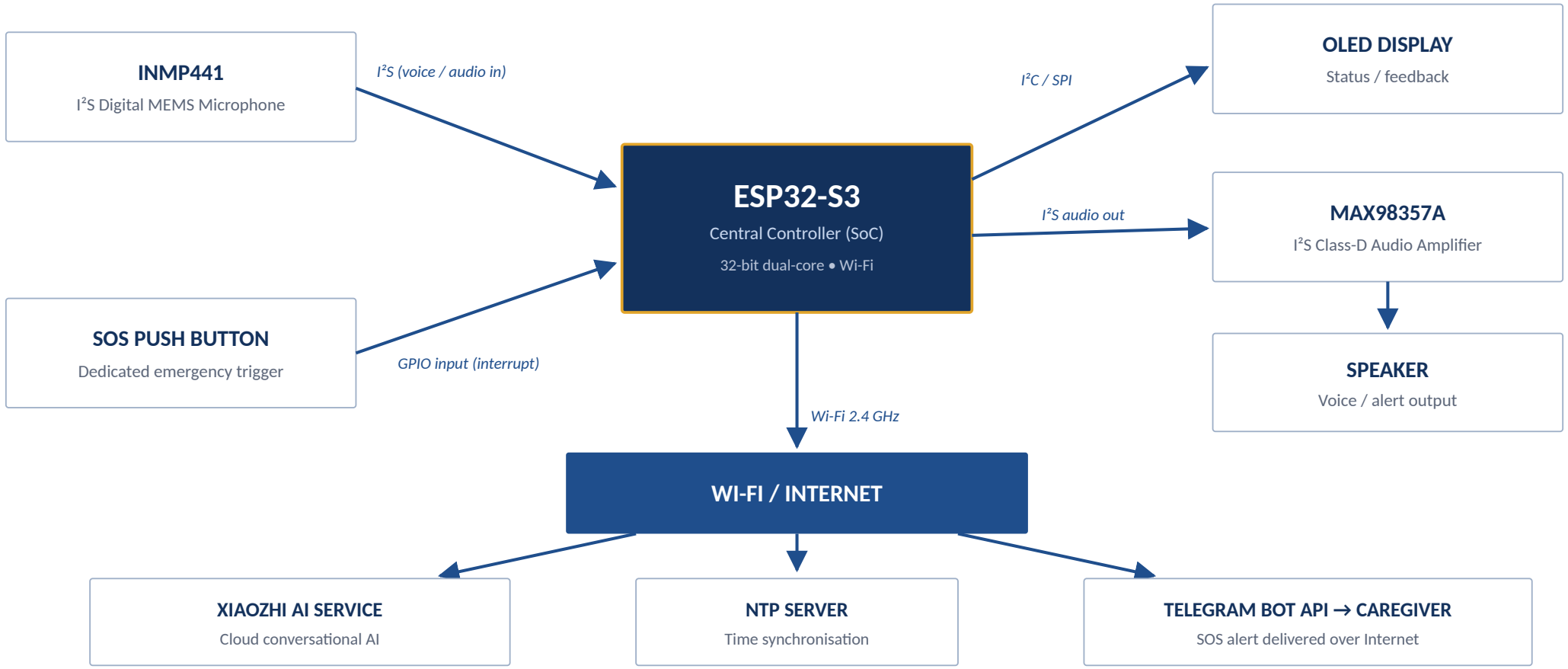
Behaviour Specification of Each Function

Function	Trigger	System Behaviour	Output
Emergency SOS	SOS button press (GPIO) or keyword “Help” / “Emergency”	Ask for confirmation → if confirmed, send message over Wi-Fi via Telegram Bot API; if not, cancel	Telegram alert to caregiver; spoken + OLED confirmation
Medicine Reminder	Internal clock matches a stored reminder time	Clock set by NTP; firmware periodically compares current time with next scheduled reminder	Speaker: “It is time to take your medicine.” OLED: medicine info
Wellness Check-in	Scheduled check-in time	Ask a wellness question → analyse spoken reply → classify Positive / Neutral / Concerning	Supportive voice reply; optional caregiver flag if concerning repeatedly
General AI Chat	User speaks a non-emergency request	Voice → text / intent classification via XiaoZhi cloud AI	Voice and display response

Scope note: the wellness function is a supportive interaction, not a medical or clinical diagnosis system. Medicine schedules are fixed in firmware for this prototype.

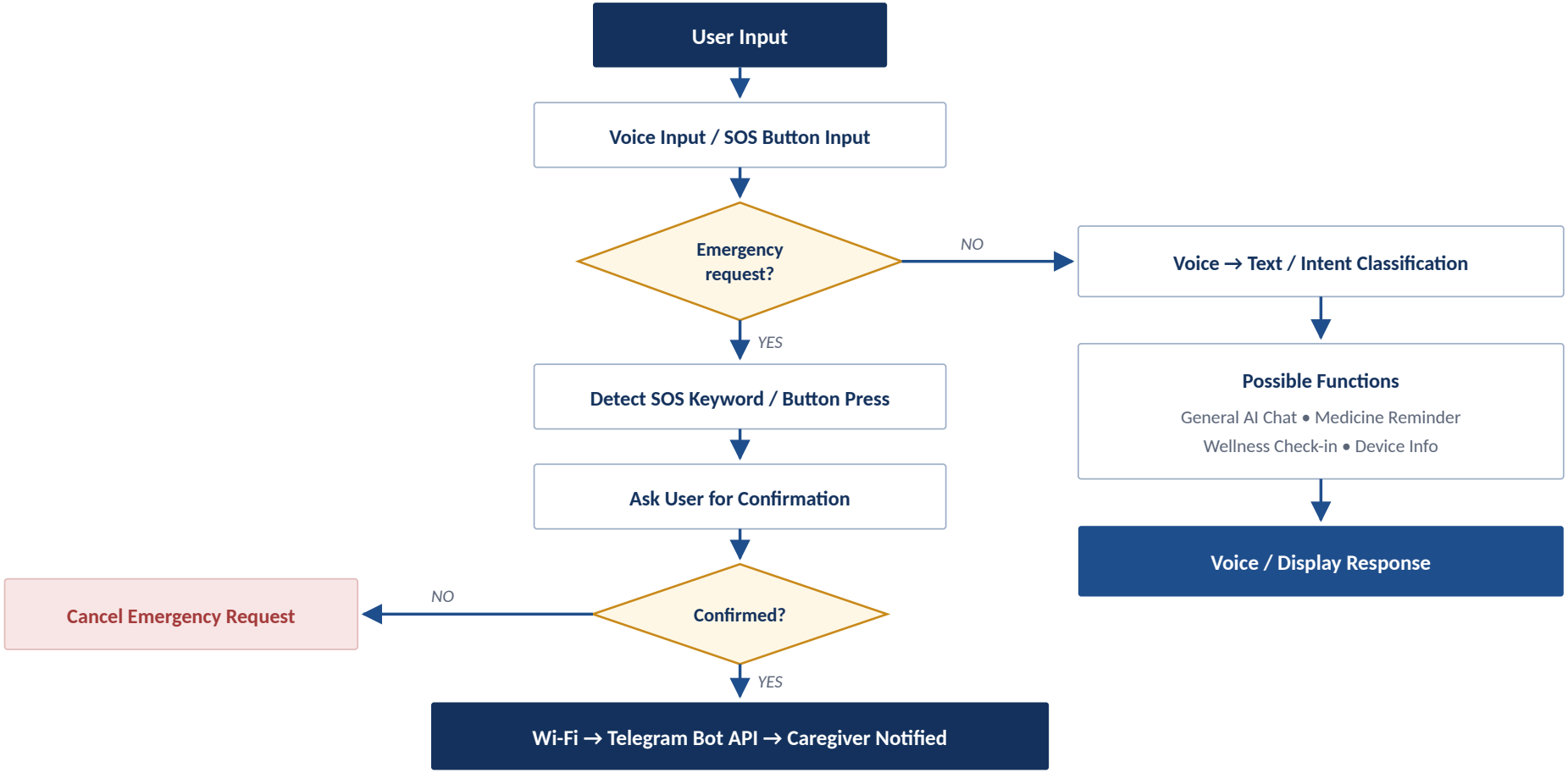
Stage 3 · Architecture

Hardware Architecture



Stage 3 · Architecture

Software Architecture — Flowchart



Stage 3 · Architecture

Firmware Architecture & RTOS Mapping

Application Tasks

Audio / AI Task mic → cloud → speaker	**SOS Handler** button / keyword → confirm	**Telegram Notifier** HTTPS to Bot API
Reminder Scheduler clock vs. schedule	**Wellness Task** scheduled check-ins	**Display Task** OLED status

XiaoZhi base firmware · Wi-Fi / NTP services

ESP-IDF RTOS kernel (FreeRTOS) + device drivers (I^2S, I^2C, GPIO)

ESP32-S3 Hardware

RTOS Kernel Service	Use in Our Firmware
Task management	Separate tasks for audio, SOS, reminders, display
Task scheduling	SOS handler given higher priority than chat
Task synchronisation	Queues / events pass SOS and reminder events between tasks
Interrupt handling	SOS button GPIO interrupt
Time management	System clock set by NTP; periodic reminder checks
Memory management	Audio buffers in SRAM

Device drivers act as the bridge between the OS services and the microphone, amplifier, OLED and button hardware.

Stage 4 • Components

Hardware Components

Component	Role in Project	Interface	Embedded System Element
ESP32-S3 Development Board	Central controller and Wi-Fi connectivity	—	System core (microcontroller / SoC) + memory
INMP441 I²S Microphone	Voice and emergency-keyword input	I²S	Input port (sensor)
SOS Push Button	Dedicated hardware emergency trigger	GPIO	Input port
MAX98357A I²S Amplifier	Digital audio output amplification	I²S	Output port (actuator driver)
Speaker	Voice response, reminders and SOS confirmation	Analog from amp	Output port (actuator)
OLED Display	Status, reminders and visual feedback	I²C / SPI	Output port
Jumper Wires + USB Power	Interconnection, programming and power	USB	Supporting subsystem / comm. interface

Stage 4 · Components

Software Components & Development Flow

Software / Services

Visual Studio Code
Firmware editing and project management

ESP-IDF
Build, flash, monitor and configure the ESP32-S3

Git + GitHub
Version control and XiaoZhi repository integration

XiaoZhi ESP32 Firmware
Base AI voice-assistant platform

Telegram Bot API
Internet-based caregiver SOS notification

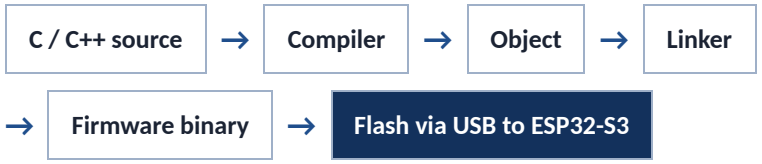
NTP
Network time synchronisation for medicine reminders

Host & Target Split

Host (PC)
VS Code + ESP-IDF toolchain compile, link and flash the firmware

Target
ESP32-S3 runs the built firmware on the device

Embedded Software Development Process



Limited observability: with no keyboard or full screen on the device, we debug using the ESP-IDF serial monitor on the host plus OLED status messages.

Functionality 1

Emergency SOS System

Two independent triggers improve accessibility and safety.

METHOD 1 — PHYSICAL SOS BUTTON

1. Elderly user presses dedicated emergency button
2. ESP32-S3 detects GPIO input
3. System requests confirmation
4. User confirms request
5. Wi-Fi connection established
6. Telegram Bot API → caregiver alerted

METHOD 2 — VOICE-BASED SOS

1. User speaks keyword: “Help” / “Emergency”
2. Emergency intent detected from speech
3. System asks for confirmation
4. User confirms request
5. Wi-Fi connection established
6. Telegram SOS message sent

The confirmation step reduces accidental or false alerts. Since the ESP32-S3 uses no cellular module, the alert travels entirely over its Wi-Fi Internet connection.

SOS ALERT: Emergency assistance requested by the user. Please check on them immediately.

Functionality 2

Medicine Reminder System

Medicine schedules are configured in firmware for the prototype; the ESP32-S3 obtains accurate time using NTP (Network Time Protocol).

1

Time Sync

ESP32-S3 connects to Wi-Fi and contacts an NTP server to obtain current date/time.

→

2

Schedule

Medicine name and reminder times are stored in firmware / configuration.

→

3

Time Match

Firmware periodically compares current time with the next scheduled reminder.

→

4

Alert

Speaker announces the reminder; OLED displays medicine information.

“It is time to take your medicine.”

The Internet is mainly needed to synchronise time initially and periodically. Between synchronisation events, the system keeps checking its internal clock to trigger reminders on schedule.

Embedded concept: a time-driven task using the RTOS time-management service and the on-chip timer — a classic periodic, soft real-time activity.

Functionality 3

Wellness Check-In — “State of Mind”

The assistant periodically interacts with the elderly user to understand their general emotional state.

Example Prompts

- **“How are you feeling today?”**
- **“How has your day been?”**
- **“Are you feeling okay?”**

Process Flow

1. Scheduled check-in triggers
2. Assistant asks wellness question
3. User responds by voice
4. Response / intent analysis
5. Classify: Positive • Neutral • Concerning
6. Provide supportive voice response

If concerning responses are detected repeatedly, the system can optionally notify or flag the situation for the caregiver.

Scope note: this is a wellness interaction, not a medical or clinical diagnosis system.

Stage 5 · System Integration

Integration & Testing Plan

Components are assembled incrementally and tested together — reliability is designed in from the start, not tested in at the end.

1
Base firmware
Integrate XiaoZhi firmware; build & flash with ESP-IDF

2
Hardware bring-up
Configure mic, amplifier, OLED and button pins

3
SOS path first
Button / keyword → confirm → Telegram message

4
Reminders
NTP sync + scheduled voice/OLED alerts

5
Wellness logic
Check-in prompts & response classification

Test Case	Procedure	Expected Result
SOS via button	Press button, confirm by voice	Caregiver receives Telegram SOS message
False-alarm rejection	Press button, decline confirmation	Request cancelled; no message sent
SOS via voice	Say “Help”, then confirm	Telegram SOS message received
Medicine reminder	Set reminder 2 minutes ahead	Voice prompt + OLED info at scheduled time

Theory → Project

Embedded Characteristics & Design Challenges

Characteristics of Embedded Computing

Characteristic	In Our Device
Sophisticated functionality	Speech interaction with cloud AI and intent detection
Real-time	Soft real-time: SOS and reminders must arrive on time to be useful
Multirate	Continuous audio streaming, per-minute reminder checks, periodic NTP sync and occasional button events run together
Cost	Set by CPU choice, memory and I/O — one SoC keeps it low
Power & energy	Always-on USB supply; heat kept low

Key Challenges & Our Approach

How much hardware? One ESP32-S3 plus I²S modules — enough for audio without extra cost.

Meeting deadlines? SOS handler runs as a high-priority task.

Upgradeability? Firmware can be rebuilt and re-flashed with new features.

Does it really work? Confirmation step and a hardware button as backup for voice.

Is it secure? Bot token and Wi-Fi credentials kept in firmware config, not shared.

Known limitation: all cloud features need Wi-Fi; no cellular fallback.

Reference / Literature Survey

Related Published Research

Author (Year)	Publication Venue	Relevance to This Project
Arnold et al. (2024)	Disabil Rehabil Assist Technol. DOI: 10.1080/17483107.2022.2065369	Scoping review of personal voice assistants for older adults; found common uses such as reminders and information access.
Kim & Choudhury (2021)	Computers in Human Behavior 124, 106914 DOI: 10.1016/j.chb.2021.106914	16-week home deployment with older adults; shows convenience, companionship and usability challenges of smart speakers.
Shade et al. (2021)	JMIR Form Res 5(9): e26361 DOI: 10.2196/26361	Feasibility study of voice-assistant medication reminders in older adults; directly motivates our reminder function.
Lersilp et al. (2020)	The Scientific World Journal DOI: 10.1155/2020/5102849	Usability-tested emergency alert device with conscious SOS triggering; supports a simple, dedicated emergency control.
Sokullu et al. (2020)	Internet of Things 11: 100239 DOI: 10.1016/j.iot.2020.100239	IoT smart-home system for elderly users, including caregiver notification and a one-button emergency mechanism.

LITERATURE GAP ADDRESSED Our prototype combines a voice-first interface, a dedicated SOS hardware trigger, Internet-based caregiver notification without a cellular module, medicine reminders and wellness check-ins in one low-cost ESP32-S3 device.

References

Full Citation Details

1. Arnold, A., Kolody, S., Comeau, A., & Cruz, A. M. (2024). What does the literature say about the use of personal voice assistants in older adults? A scoping review. *Disability and Rehabilitation: Assistive Technology, 19*(1), 100–111. DOI: 10.1080/17483107.2022.2065369.

2. Kim, S., & Choudhury, A. (2021). Exploring older adults' perception and use of smart speaker-based voice assistants: A longitudinal study. *Computers in Human Behavior, 124*, 106914. DOI: 10.1016/j.chb.2021.106914.

3. Shade, M., Rector, K., & Kupzyk, K. (2021). Voice Assistant Reminders and the Latency of Scheduled Medication Use in Older Adults With Pain: Descriptive Feasibility Study. *JMIR Formative Research*, *5*(9), e26361. DOI: 10.2196/26361.

4. Lersilp, S., Putthinoi, S., Lerttrakarnnon, P., & Silsupadol, P. (2020). Development and Usability Testing of an Emergency Alert Device for Elderly People and People with Disabilities. *The Scientific World Journal*, *2020*, 5102849. DOI: 10.1155/2020/5102849.

5. Sokullu, R., Akkaş, M. A., & Demir, E. (2020). IoT supported smart home for the elderly. *Internet of Things*, *11*, 100239. DOI: 10.1016/j.iot.2020.100239.

6. Course material: G. Boopalan, *Embedded Systems Architecture*, Module 1 lecture notes, DCE, SENSE, VIT.

Conclusion

Summary & Next Steps

What we presented

- **Core prototype:** ESP32-S3 + microphone + speaker/amplifier + OLED + Wi-Fi + dedicated SOS button
- **Differentiator:** not just a chatbot — adds elderly-care workflows for emergency response, reminders and proactive wellness
- **Method:** each design stage — requirements, specification, architecture, components, integration — applied to our device

Next implementation phase

- Integrate XiaoZhi base firmware and configure hardware
- Implement and demonstrate the SOS path first: button / keyword → confirmation → Telegram message received by caregiver
- Then add medicine reminders and wellness logic
- Refine requirements bottom-up from test results

Our goal: a clear, technically feasible, low-cost embedded device that helps elderly users live independently and keeps caregivers informed.

Thank You

Questions & Discussion

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